

Data Cleaning in Machine Learning

Complete Notes

1. What is Data Cleaning?

Data Cleaning is the process of identifying and correcting or removing incorrect, incomplete, inconsistent, duplicate, and irrelevant data before using the dataset for Machine Learning.

Example

Name	Age	Gender	Salary
Ravi	25	Male	30000
Priya	NaN	Female	40000
Arun	30	M	35000
Ravi	25	Male	30000
Kumar	200	Male	-5000

Problems:

- Age = NaN → Missing value
- M vs Male → Inconsistent category
- Ravi appears twice → Duplicate
- Age = 200 → Possible invalid/outlier value
- Salary = -5000 → Invalid value

2. Why is Data Cleaning Important?

Machine Learning algorithms learn patterns from data.

If the input data is poor, the model may learn incorrect patterns.

Poor-quality data → Poor model performance

- Improve data quality
- Reduce errors
- Handle missing information
- Remove duplicate records
- Handle outliers
- Maintain consistency
- Improve model accuracy
- Improve model reliability

A common idea is:

Garbage In → Garbage Out

If incorrect data is given to an ML model, the output can also be unreliable.

3. Main Steps in Data Cleaning

A typical data-cleaning process is:



4. Understanding the Dataset

Before cleaning, we should understand the dataset.

- Number of rows
- Number of columns
- Column names
- Data types
- Missing values
- Unique values
- Numerical columns
- Categorical columns

Important questions

- What does this column represent?
- Is the data type correct?

- Can this column contain missing values?
- Are the values valid?
- Is the column useful for ML?

5. Handling Missing Values

A missing value means information is not available for a particular record.

Name	Age	Salary
Arun	25	30000
Priya	NaN	40000
Kumar	30	NaN

Here:

- Priya's age is missing.
- Kumar's salary is missing.

Missing values may appear as:

NaN

NULL

None

?

Unknown

N/A

5.1 Types of Missing-Value Handling

Method 1: Remove rows

If only a small number of records contain missing values, we may remove those rows.

Example:

1000 rows

10 rows have missing values

Removing 10 rows may be acceptable depending on the problem.

But if:

1000 rows

600 rows have missing values

removing them would result in significant data loss.

Method 2: Fill with Mean

For numerical data:

10, 20, 30, NaN, 40

Mean:

$$(10 + 20 + 30 + 40) / 4 = 25$$

Missing value can be replaced with:

25

Mean is sensitive to outliers.

Method 3: Fill with Median

Example:

10, 20, 30, NaN, 40

Median:

30

Median is often preferred when the numerical column contains outliers or is highly skewed.

Method 4: Fill with Mode

Mode is the most frequently occurring value.

Example:

Male

Female

Male

NaN

Male

Mode:

Male

The missing value can be replaced with Male.

Mode is commonly used for categorical data.

Method 5: Forward Fill / Backward Fill

Sometimes missing values can be filled using nearby observations.

Forward fill:

10

20

NaN

40

becomes:

10

20

20

40

Backward fill:

10

20

NaN

40

becomes:

10

20

40

40

These methods are particularly useful for certain time-series datasets.

6. Duplicate Data

A duplicate means the same record appears more than once.

ID	Name	Age
101	Ravi	25
102	Priya	28
101	Ravi	25

The Ravi record is duplicated.

Duplicates can:

- Increase dataset size incorrectly
- Bias the model
- Affect statistical calculations

Therefore, duplicates should be identified and handled appropriately.

7. Incorrect Data Types

A column may contain values in an inappropriate format.

Example:

Age
25
30
35
40

Age should normally be numerical.

But sometimes it may be stored as:

"25"

"30"

"35"

"40"

as strings.

Similarly:

"2026-01-10"

may need to be converted into a proper date/time format.

Common data types

- Integer
- Float
- String/Object
- Boolean
- Date/Time
- Categorical

Correct data types make analysis and preprocessing easier.

8. Inconsistent Data

Data can contain different representations of the same value.

Example:

Male

male

M

MALE

These may all represent the same category.

Another example:

Chennai

chennai

CHENNAI

Madras

Depending on the context, these may need to be standardized.

Solution

Convert values into a consistent format.

Male

Male

Male

Male

9. Spelling Errors

Categorical data can contain spelling mistakes.

Example:

Chennai
Chennai
Chenai
Chennnai

Chenai and Chennnai may be spelling errors.

These errors can create unnecessary categories.

Correct:

Chennai
Chennai
Chennai
Chennai

10. Invalid Values

Some values may be logically impossible or invalid.

Age
25
32
-5
200
28

An age of -5 is invalid.

An age of 200 is likely invalid for a normal human-age dataset.

Another example:

Salary = -10000

If negative salary is not meaningful for the business problem, it should be investigated.

Important: Don't automatically delete unusual values. First understand the domain.

11. Outliers

An outlier is a value that is unusually far from the majority of observations.

Example:

20
22
21
23

24
22
25
150

Here, 150 may be an outlier.

Outliers can occur because of:

- Data-entry errors
- Measurement errors
- Genuine unusual observations
- Fraud or abnormal behavior
- Rare events

11.1 Handling Outliers

- Remove the outlier — Use when the value is clearly an error.
- Replace the value — If the value is incorrect and the correct value can be determined.
- Cap the value — Limit extremely high/low values.
- Use transformation — For example, logarithmic transformation may reduce the influence of extreme values.
- Keep the outlier — If it represents a genuine observation.

Important: An outlier is not automatically an error.

12. Detecting Outliers

IQR Method

IQR means:

Interquartile Range

$$\text{IQR} = Q3 - Q1$$

Lower limit:

$$Q1 - 1.5 \times \text{IQR}$$

Upper limit:

$$Q3 + 1.5 \times \text{IQR}$$

Values outside these limits can be considered potential outliers.

Z-Score

Z-score measures how far a value is from the mean in terms of standard deviations.

$$Z = (X - \text{Mean}) / \text{Standard Deviation}$$

A commonly used rule is that values with an absolute Z-score greater than about 3 may be potential outliers, although the appropriate threshold depends on the problem and distribution.

13. Irrelevant Data

Some columns may not contribute useful information to the ML problem.

Customer_ID	Name	Age	Income	Purchased
101	Ravi	25	30000	Yes

If Customer_ID is simply an identifier, it may not provide useful predictive information.

Similarly, a column containing completely irrelevant information may be removed.

Important

Don't remove a column simply because it looks unnecessary.

First understand:

- Its purpose
- Its relationship to the target
- Whether it contains useful information
- Whether using it would cause data leakage

14. Handling Categorical Data

Machine Learning algorithms generally require numerical representations.

Example:

Gender

Male

Female

Male

Female

Possible encoding:

Male → 0

Female → 1

For categories without an inherent order, one-hot encoding is commonly used.

Chennai	Bangalore	Mumbai
1	0	0
0	1	0
0	0	1

This is generally part of data preprocessing, although it is often discussed alongside data cleaning.

15. Handling Text Data

Text columns may contain:

- Extra spaces
- Different capitalization
- Special characters
- Spelling variations

Example:

" Chennai"

"CHENNAI"

"chennai "

These can be standardized where appropriate.

For NLP projects, additional preprocessing may include:

- Tokenization
- Lowercasing
- Stopword handling
- Removing unwanted characters
- Stemming
- Lemmatization

These are more specifically text preprocessing steps.

16. Handling Date and Time Data

Dates may appear in different formats:

25-09-2026

2026-09-25

09/25/2026

They may represent the same date but use different formats.

Dates should be standardized and converted into a proper date/time representation.

From a date, we can also derive:

- Year
- Month
- Day
- Day of week
- Hour
- Quarter

These derived variables can later be used for ML or analytics.

17. Data Validation

After cleaning, we should verify the dataset again.

Check missing values

Are important missing values still present?

Duplicates

Are unwanted duplicate records removed?

Data types

Are columns stored correctly?

Values

Are values within reasonable ranges?

Categories

Are category names consistent?

Dataset size

Did we accidentally remove too many rows?

18. Data Cleaning vs Data Preprocessing

Data Cleaning	Data Preprocessing
Fix missing values	Transform data for ML
Remove duplicates	Encode categorical variables
Fix inconsistent values	Scale numerical features
Correct invalid data	Normalize/standardize
Handle problematic outliers	Feature transformation
Remove irrelevant data	Prepare model-ready features

These terms are related but not exactly the same.

A simplified ML workflow is:

Raw Data → Data Cleaning → Data Preprocessing → Feature Engineering → Train/Test Split → Model Training → Evaluation

Some workflows combine these stages under the broader term data preprocessing.

19. Common Data Cleaning Checklist

Dataset

- Understand rows and columns
- Understand each feature
- Identify target variable
- Check data types

Missing Data

- Identify missing values
- Understand why values are missing
- Remove or impute appropriately

Duplicates

- Identify duplicate records
- Remove duplicates when appropriate

Consistency

- Standardize categorical values
- Fix spelling errors
- Remove unnecessary spaces
- Standardize formats

Invalid Data

- Check impossible values
- Check incorrect ranges
- Check incorrect units

Outliers

- Detect potential outliers
- Investigate their cause
- Remove, transform, cap, or retain appropriately

Columns

- Remove irrelevant columns
- Check for leakage
- Check whether identifiers are useful

Final Validation

- Check missing values again
- Check duplicates again
- Check data types
- Check value ranges
- Confirm dataset is ready for preprocessing

20. Simple Example

Suppose we have:

Age	Gender	Salary
25	Male	30000
NaN	Female	40000
30	M	35000
25	Male	30000
-5	Male	-10000

Step 1 — Missing value

Age = NaN

Investigate and impute appropriately.

Step 2 — Duplicate

The first and fourth rows are duplicates.

Remove the unwanted duplicate.

Step 3 — Inconsistent category

Male

M

Standardize them if they represent the same category.

Step 4 — Invalid values

Age = -5

Salary = -10000

Investigate and correct/remove according to the domain rules.

Final result

The dataset becomes more consistent and suitable for further preprocessing and ML.

21. Important Points to Remember

- Never clean data blindly.
- Understand the meaning of each column before modifying it.
- Missing values don't always mean the row should be deleted.
- An outlier is not necessarily an error.
- Don't remove columns without understanding their purpose.
- Avoid data leakage during cleaning and preprocessing.
- Apply transformations using information available from the training data when building ML pipelines.
- Always validate the dataset after cleaning.
- Document important cleaning decisions.
- Good data quality is essential for reliable ML models.

One-line definition for students

Data Cleaning is the process of identifying, correcting, removing, or handling inaccurate, incomplete, inconsistent, duplicate, and irrelevant data before using it for Machine Learning.